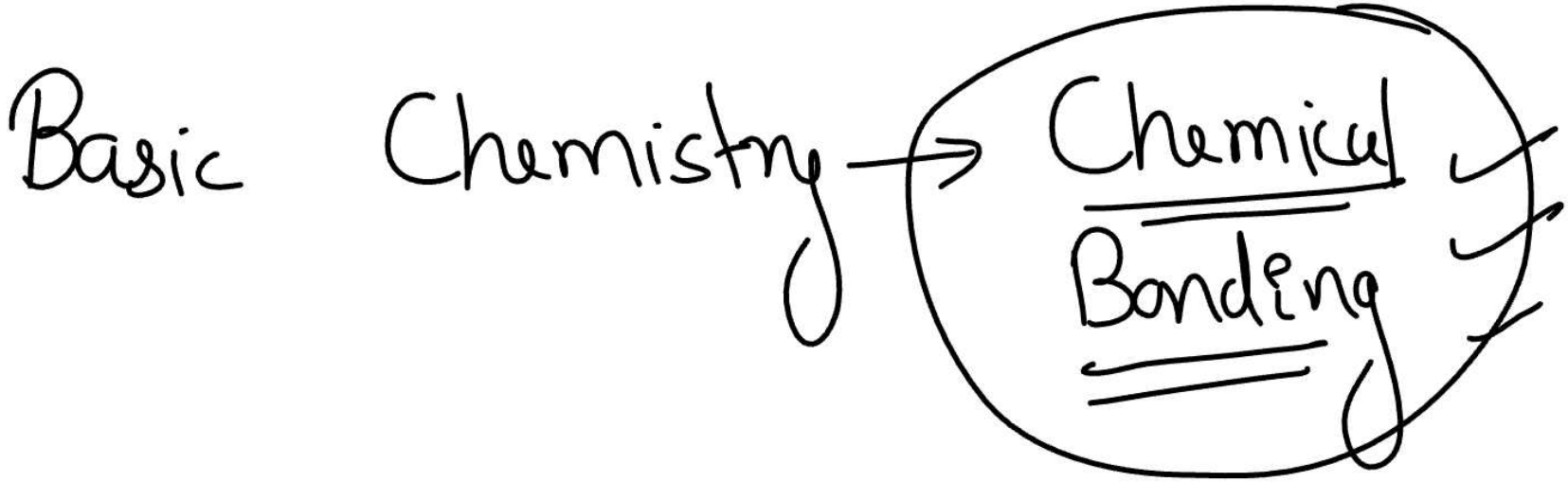
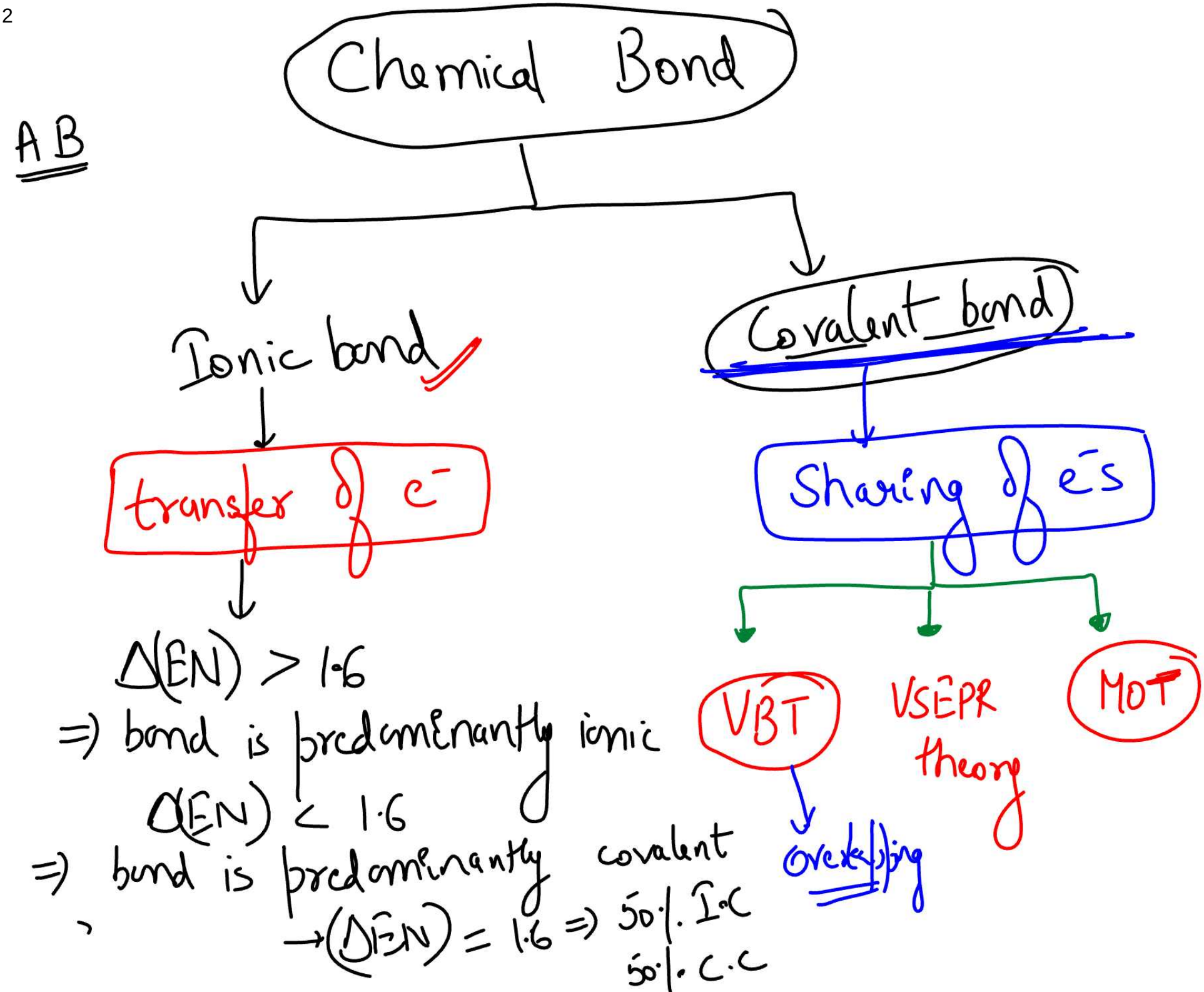
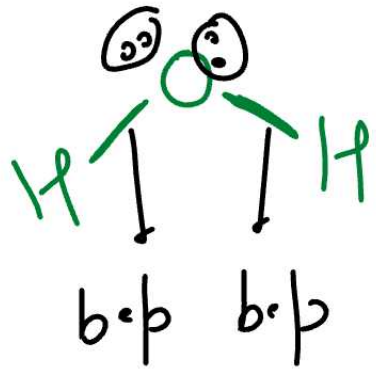
Coordination Compounds

Tue May 05 2026




Chemical Bond
AB
Ionic bond
Covalent bond
transfer of e⁻
Sharing of e⁻s
Δ(EN) > 1.6
⇒ bond is predominantly ionic
Δ(EN) < 1.6
⇒ bond is predominantly covalent
→ (ΔEN) = 1.6 ⇒ 50% I.C
50% C.C
VBT
VSEPR theory
MOT
overlapping

H_2O →



Order of repulsion →

l.p - l.p > l.p - b.p > b.p - b.p

Valence Bond Theory

Bond is formed by overlapping of valence orbitals of bonding atoms.

Strength of bond $\propto$ extent of overlapping

Overlapping

- Head on overlap ⇓ σ-bond
- lateral overlap ⇓ π-bond

Internuclear axis →



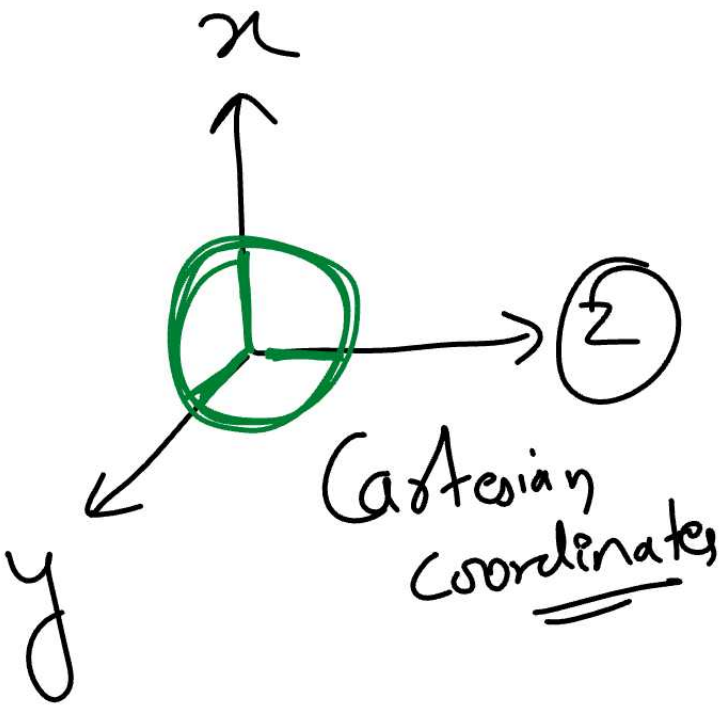
Orbitals

present along z-axis

⊥ to z-axis

s, p_z, d_{z^2} → Head on overlap

all other orbitals → lateral overlap.



Cartesian coordinates

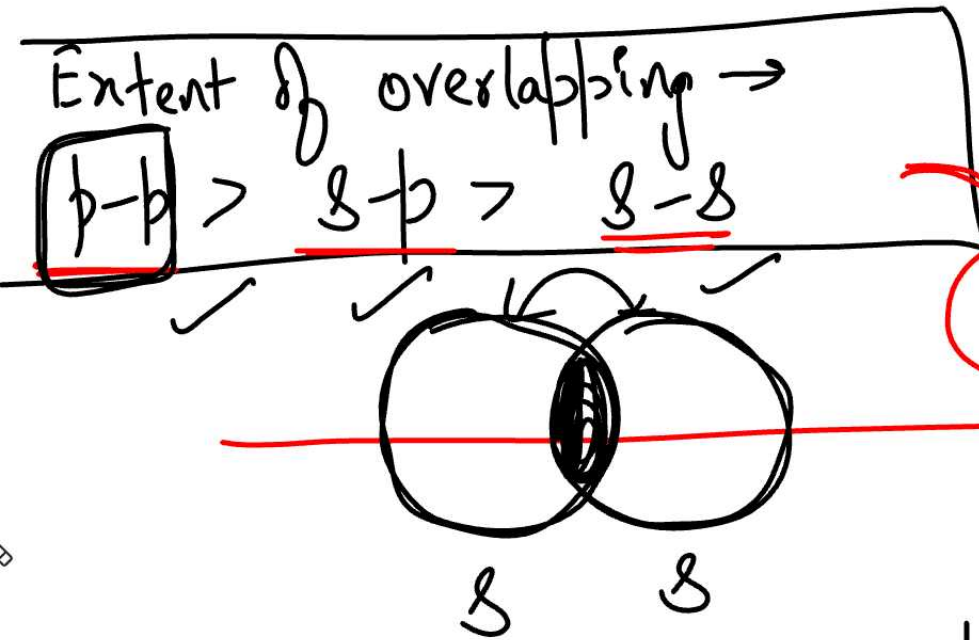
Head on overlap → It is overlapping of orbitals along the internuclear axis.

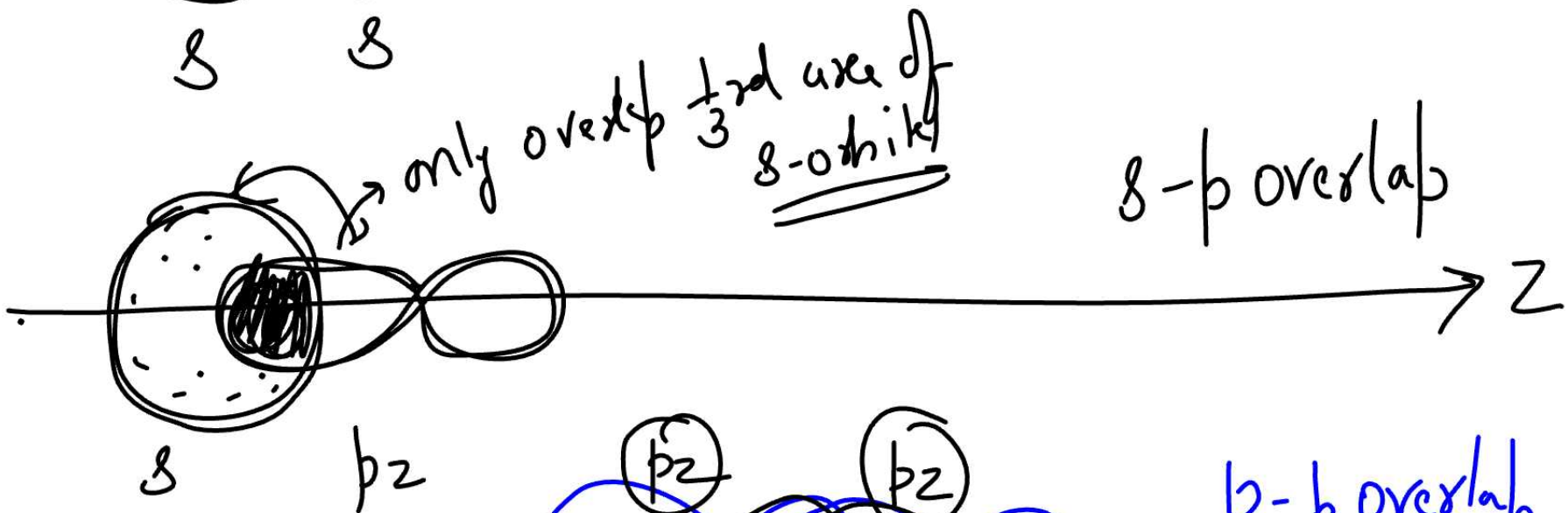
or σ-overlap.

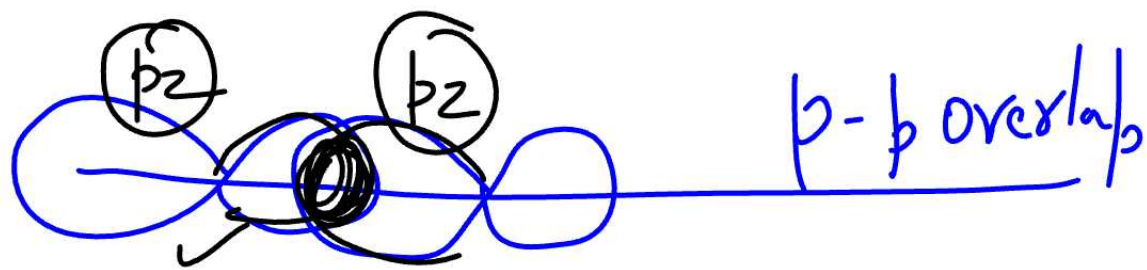
s, p_z, d_{z^2}

Extent of overlapping → p-p > s-p > s-s

σ bond

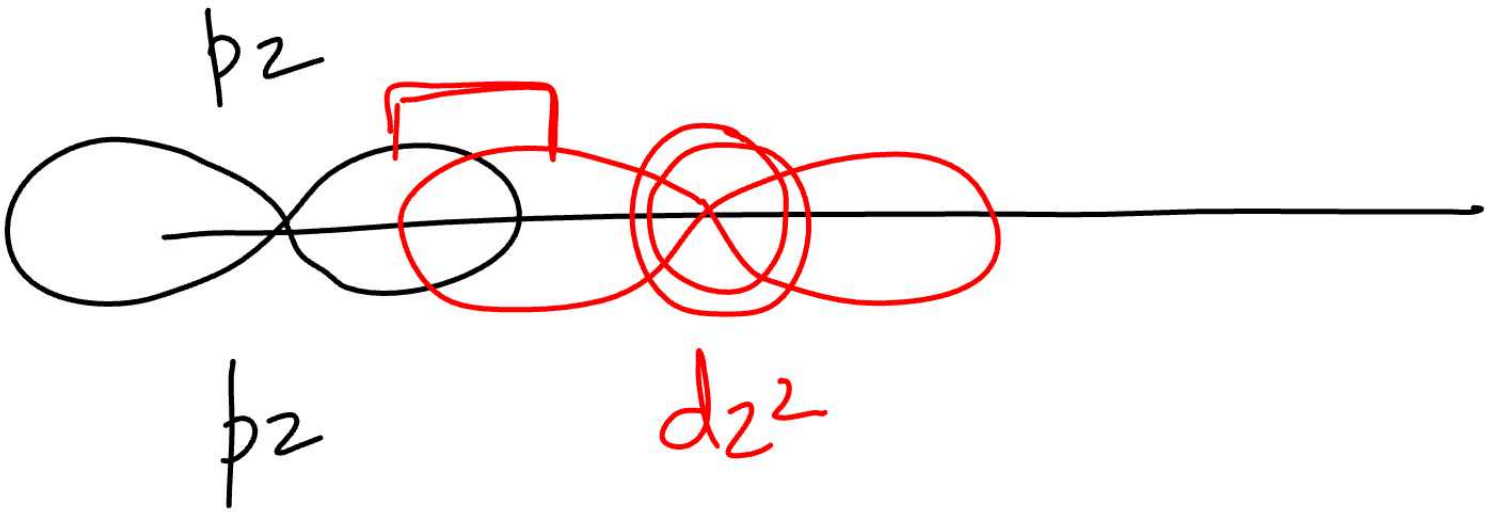




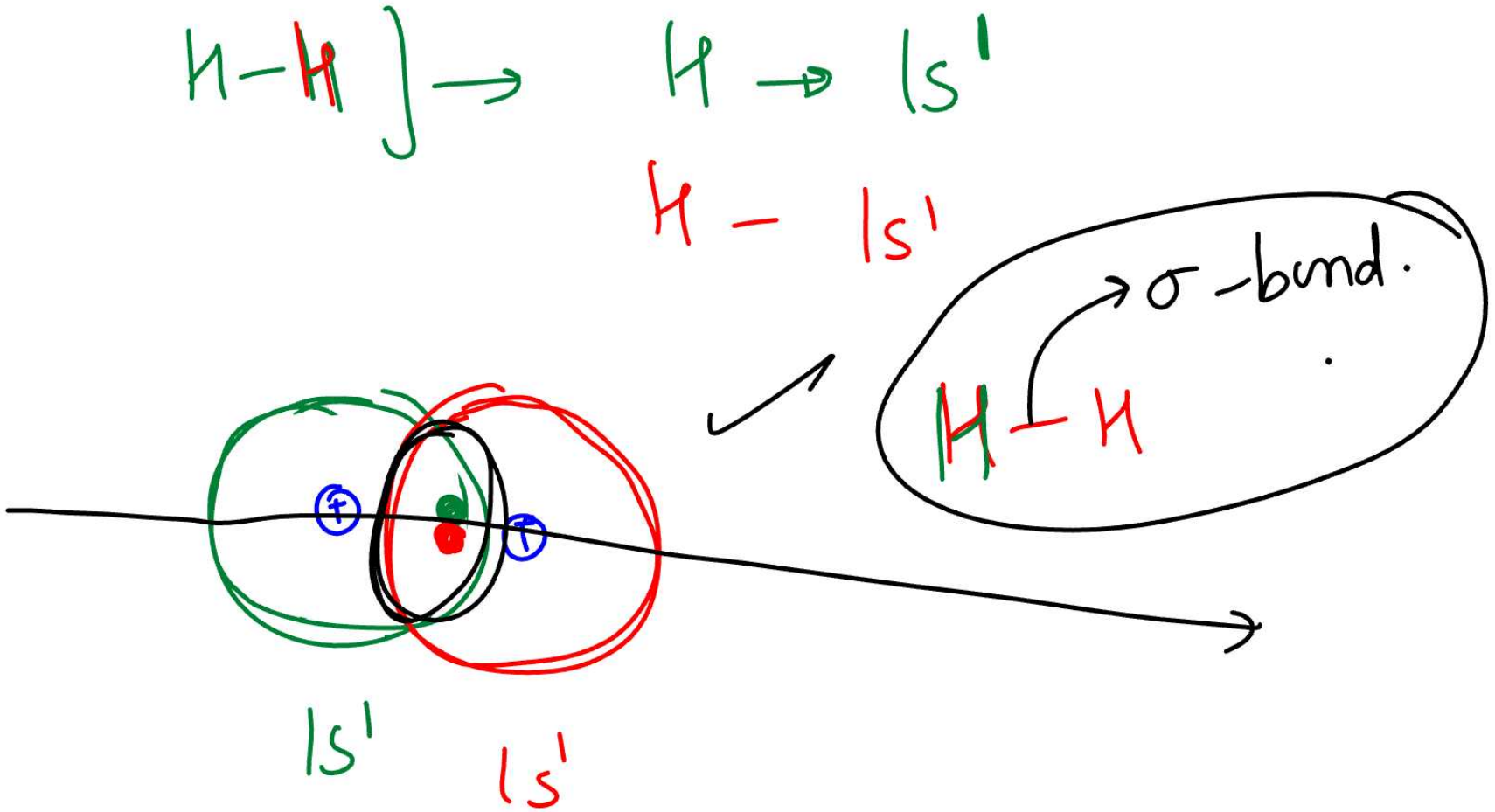


About d_{z^2}

& p d

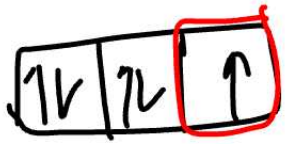


p-d overlap


H-H → H → 1s¹
H - 1s¹
σ-bond.
H-H
1s¹
1s¹

F_2

$F \rightarrow 1s^2 2s^2 2p^5$

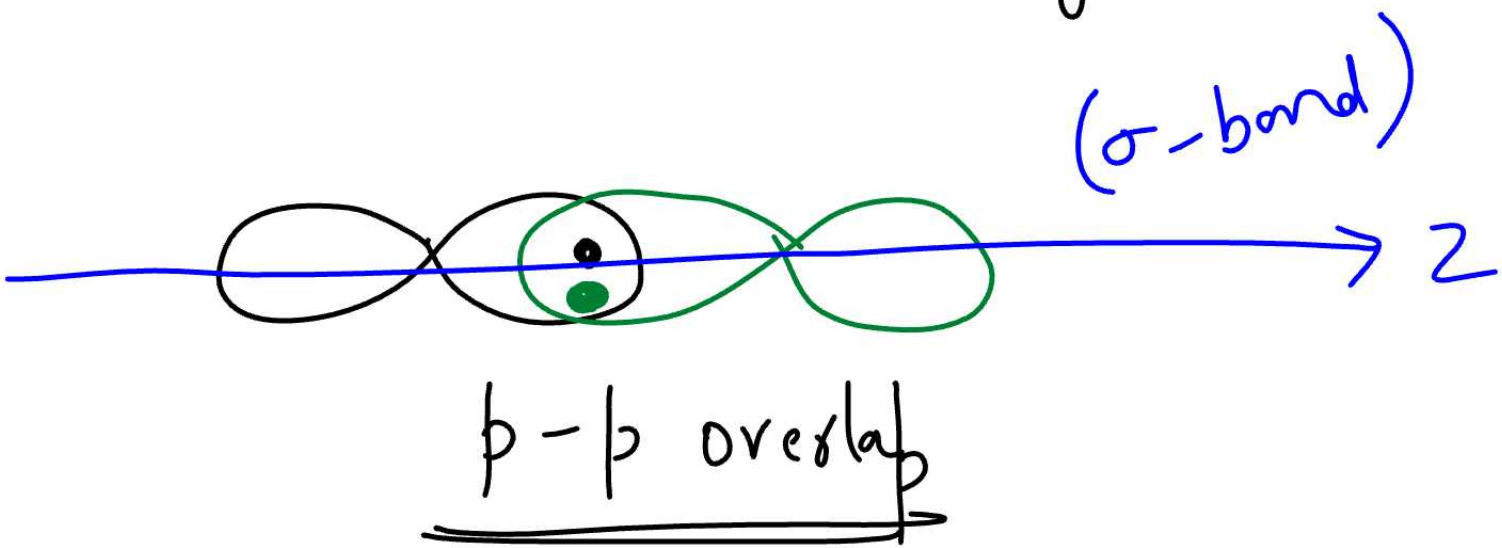


p_x p_y p_z

$F \rightarrow 1s^2 2s^2 2p^5$



p_x p_y p_z



p - p overlap

Types of overlapping

Positive overlap

or additive or constructive overlap

When orbitals overlap through lobes of same sign.

(Bonding situation)

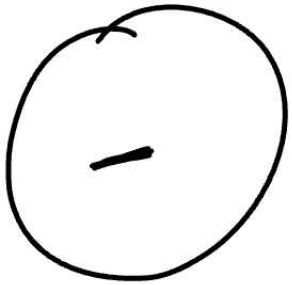
Negative overlap

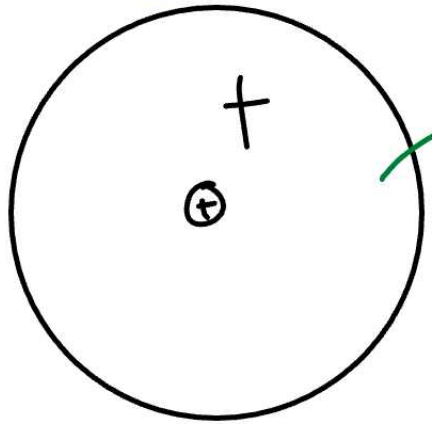
When orbitals overlap through lobes of different signs.

(Antibonding situation)

Shapes of orbitals

s-orbital

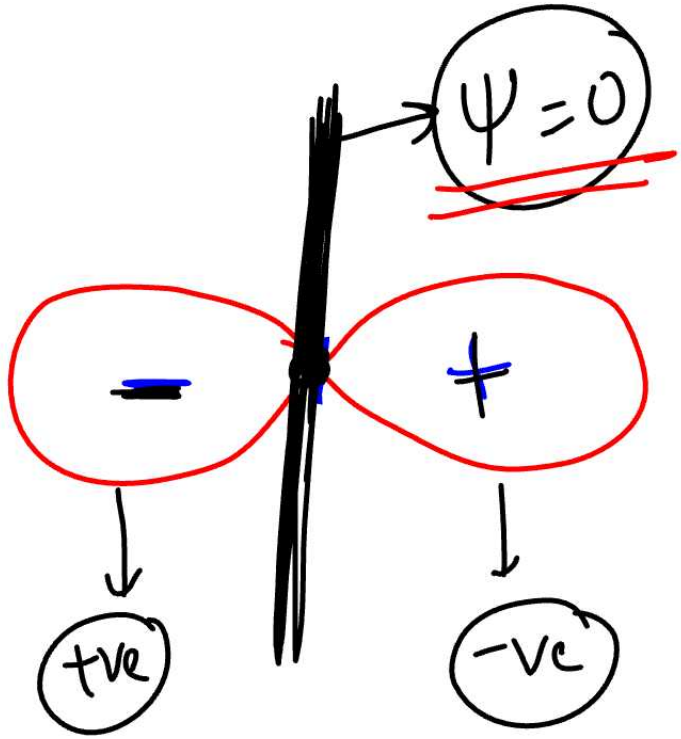


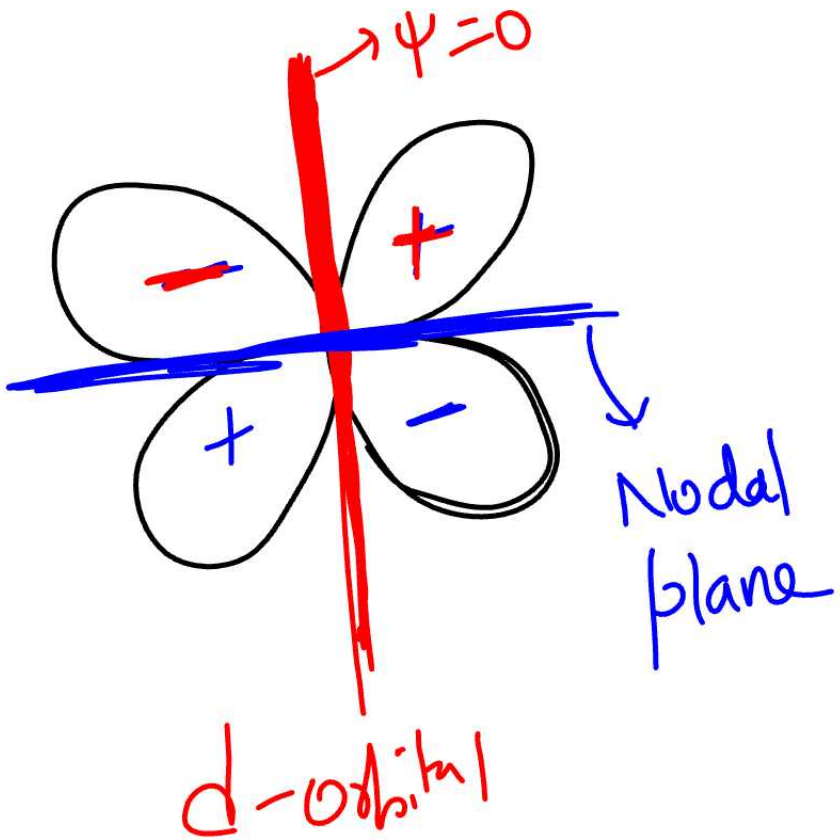


ψ → represent an e⁻ in an orbital

+ve value

p-orbital

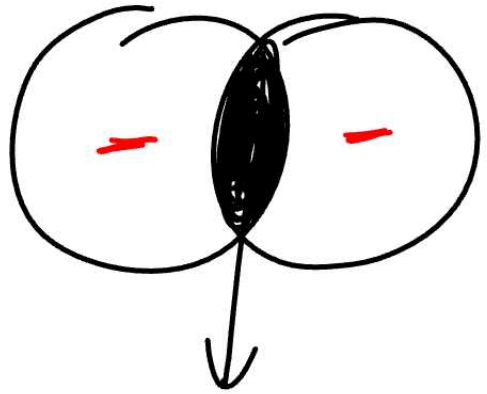




ψ = non-zero
⇒ e^- exists at that point.

$\psi = 0$ ⇒ e^- doesn't exist.

Nodal plane → A plane passing through orbital on which e^- density = 0 ie $\psi = 0$ or $\psi^2 = 0$



→ +ve overlap.

ψ is increased

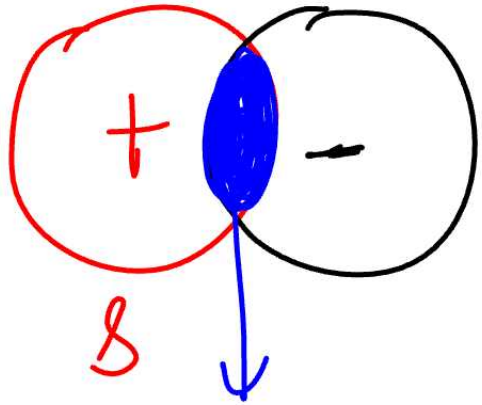
$\Rightarrow$ e⁻ density increases

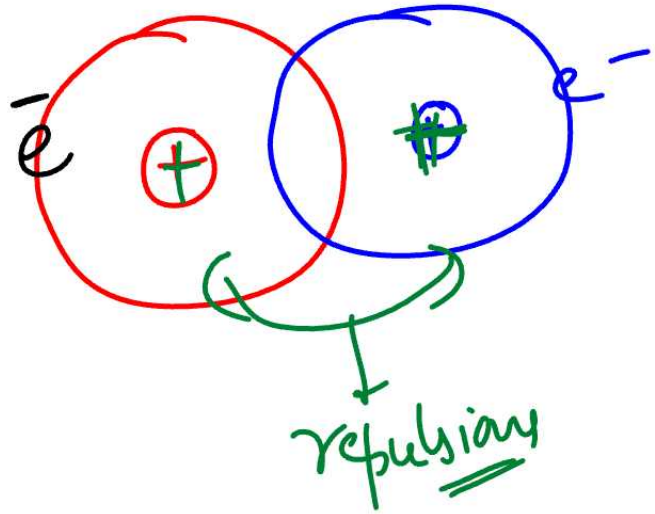
$\Rightarrow$ Bonding thing

(+1) + (+1) = (+2)

ψ = (+2)

+ve overlap leads to bond formation



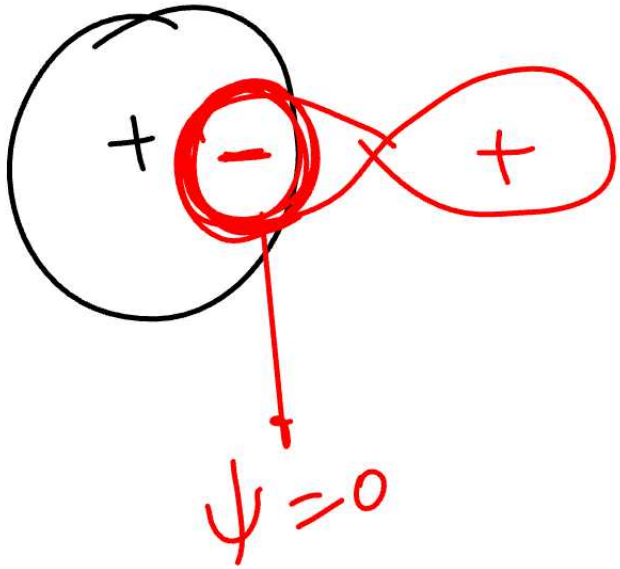


$\psi = 0$

$\Rightarrow$ No e^- in b/w the two atoms

$\Rightarrow$ Antibonding thing

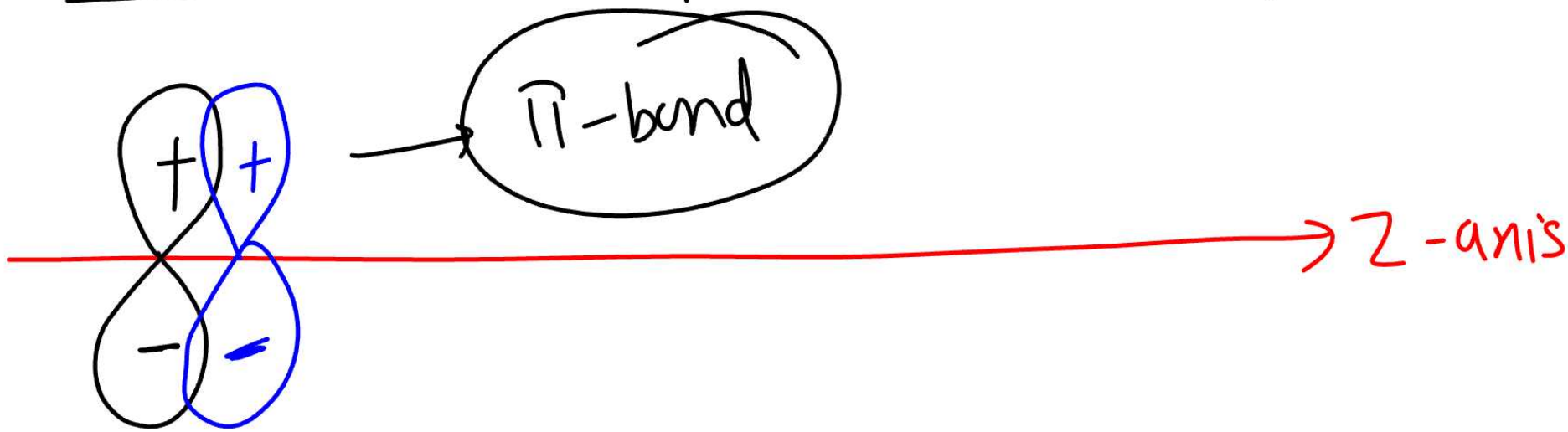
-ve overlap leads to repulsion b/w two atoms & it makes them unstable.



→ -ve overlap

No bond

Lateral overlap or π-overlap

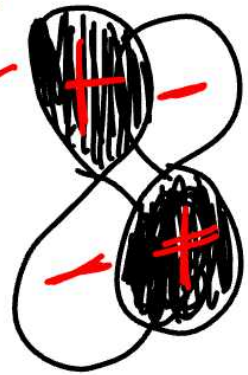


π-bond is weaker than σ-bonds.

π bonds cannot exist alone.

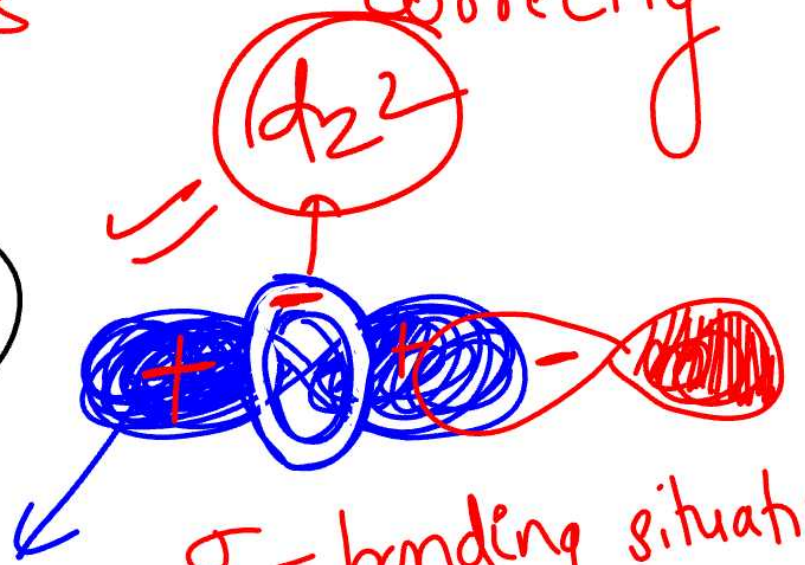
Which of these is matched?

a)



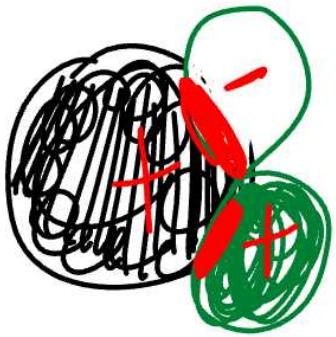
anti-Bonding situation

b) correctly



σ-bonding situation

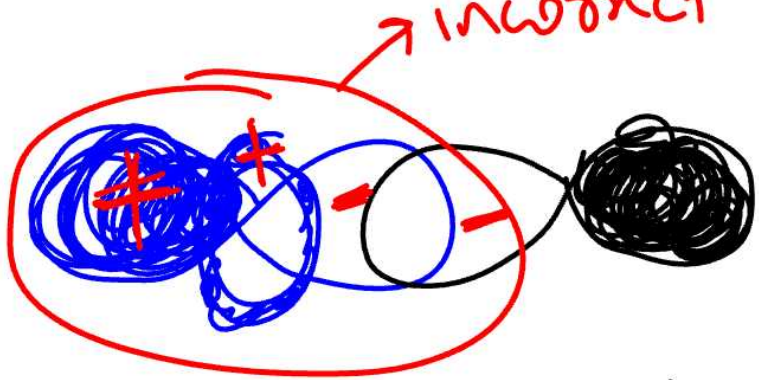
c)



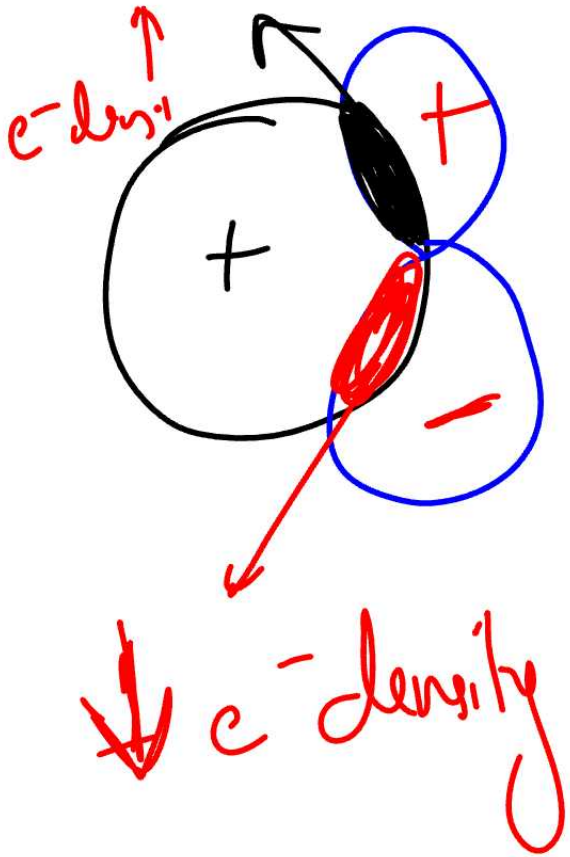
Zero overlap

Π-bonding situation

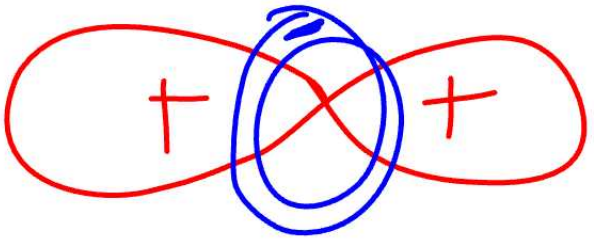
d) incorrect



σ-bonding situation


e⁻ density
+
+
−

e⁻ density


+
+

How to draw structures of molecules →

→ Select the central atom (least EN except H)

→ Complete the valency of side atoms

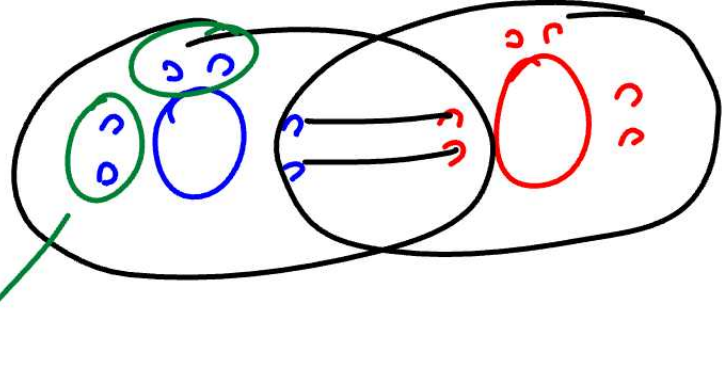
→ Type of side atoms → No. of bonds formed.

(i) H-atoms always forms 1 bond & no lop.

(ii) Halogens = (F, Cl, Br, I) forms 1 bond & 3 lops

(iii) O-atom (6 valence e⁻s) forms 2 bonds & 2 lops.

(iv) N-atoms (5 valence e⁻s) forms 3 bonds & 1 lop.



2 l.p
on each
O-atom

Total No. of l.ps on central atom in SNF_3, SO_2, CO_2

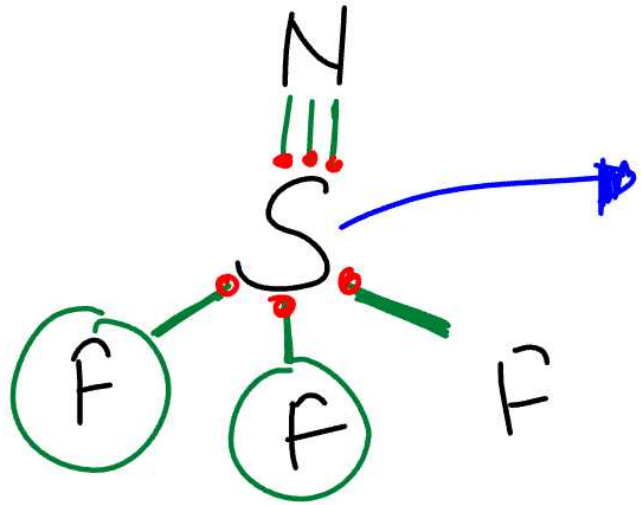
a) 1

b) 2

c) 3

d) 0

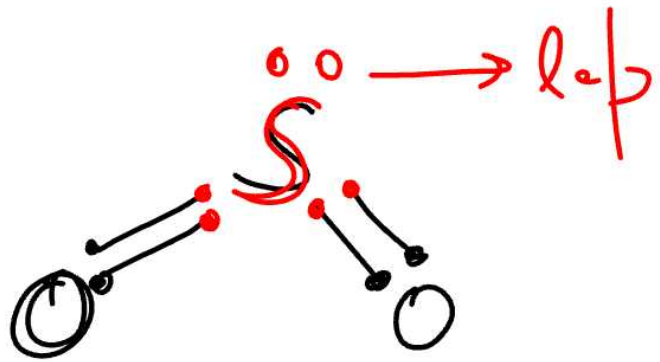
SNF_3



Valence $e^- = 6$

No l.p

SO_2



1 l.p on central atom

CO_2

O = C = O No l.p

Total l.ps = 0 + 1 + 0 = 1